Avian Population Estimation

using Passive Acoustic Monitoring

Challenge: Darukaa.earth / BioDCASE 2026 Bird Counting

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Executive Summary

This report outlines a production-oriented machine learning platform engineered to estimate bird populations from passive acoustic recordings. Built for the BioDCASE 2026 Bird Counting Challenge[2], this system moves beyond simple acoustic detection to deliver precise integer population counts.

While previous baselines for this dataset struggled with overfitting—often artificially restricting models to a single feature and yielding a Mean Absolute Error (MAE) of 45.32 without statistical significance ($p=0.385$)[1]—this pipeline successfully harnesses high-dimensional data. By combining classical digital signal processing (DSP), ecoacoustic feature engineering, and deep audio embeddings (BirdNET & PANNs), the final system achieves a **Best MAE of 1.704**. The platform is fully containerized via Docker and deployed as a FastAPI REST API, designed to satisfy the functional requirements of the challenge specification for a scalable, end-to-end ML solution.

1. Challenge & Business Requirements

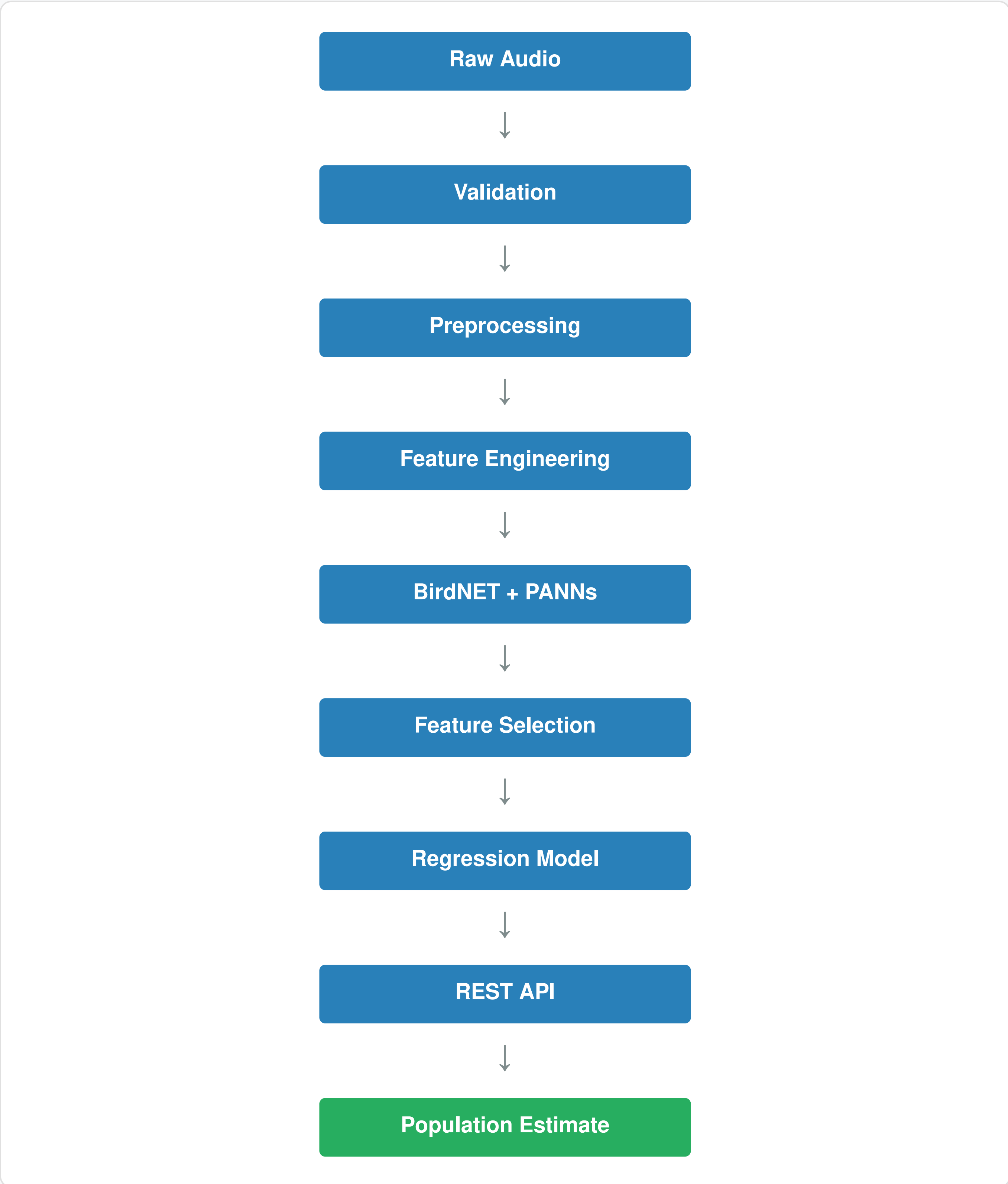
The core objective of the challenge is to process short passive acoustic recordings (3-second WAV files at 48 kHz) and estimate the population size of target species—specifically the Greater Flamingo, Red-billed Quelea, and Hadada Ibis—for each aviary[2].

The primary complexities of this task include:

- **Overlapping Vocalizations:** Handling multi-species acoustic environments.
- **Population Scale Differences:** Generalizing across species with vastly different true population sizes.
- **The Abundance Constraint:** Transitioning from binary species detection (is the bird present?) to continuous regression (how many birds are calling?).

2. Project Architecture

The prediction workflow follows a modular, end-to-end architecture transitioning from raw audio validation through to JSON-based REST API population estimation.



System Components

Component	Responsibility
Data Validation	Validate audio files and metadata
Preprocessing	Normalize and clean recordings
Feature Engineering	Extract DSP and ecoacoustic features
Embedding Extraction	Generate BirdNET and PANNs embeddings
Model Training	Train and benchmark regression models
Inference Engine	Predict bird population counts
REST API	Serve predictions via FastAPI
Testing	Automated unit and integration tests

3. Project Statistics & Technology Stack

Project Statistics

Python Modules	40+
Regression Models	19
Feature Dimensions	139
REST API Endpoints	5
Automated Tests	31
Containerization	Docker
Documentation	Complete

Technology Stack

- **Programming Language:** Python 3.10
- **Machine Learning:** Scikit-learn, CatBoost, XGBoost
- **Audio Processing:** Librosa, SoundFile, NumPy, SciPy
- **Backend:** FastAPI, Uvicorn
- **Deployment:** Docker, Docker Compose
- **Testing:** Pytest

4. Engineering Challenges

Developing a reliable estimator for the BioDCASE dataset required overcoming several significant technical hurdles:

- **Preventing data leakage** across recording locations (e.g., ensuring models didn't memorize unique room acoustics of specific aviaries).
- **Managing high-dimensional** BirdNET and PANN embeddings on limited compute constraints.
- **Designing a reproducible** feature extraction pipeline robust to environmental background noise.
- **Creating a production-ready** inference API capable of serving real-time predictions.
- **Benchmarking multiple regression algorithms** using identical, rigorous validation strategies.

5. Advanced Feature Engineering

Instead of abandoning deep learning embeddings due to the risk of learning room acoustics over true bird populations[1], this pipeline extracts a comprehensive **139-dimensional feature space** that captures both macro and micro acoustic behaviors.

- **Spectral & Temporal Features:** Capturing the physical characteristics of the sound waves.
- **Ecoacoustic Indices:** Measuring acoustic complexity, diversity, and evenness across the recordings.
- **MFCC Features:** Mel-frequency cepstral coefficients for mimicking human auditory perception.
- **Deep Audio Embeddings:** Leveraging pretrained BirdNET and PANNs models to capture high-level semantic audio representations.

By applying aggressive feature selection and dimensionality reduction, the system identifies the exact subset of these 139 features that generalize across different recording environments, completely bypassing the limitations of single-feature heuristic models.

6. Model Evaluation & Benchmarking

To ensure the models learned true biological relationships rather than memorizing site-specific acoustic anomalies, the platform evaluated **19 distinct regression algorithms** using **Group K-Fold Cross Validation**. This leakage-free validation strategy explicitly prevents data from the same location and time period from bleeding into the testing folds.

Overall Benchmark Results (Top Models)

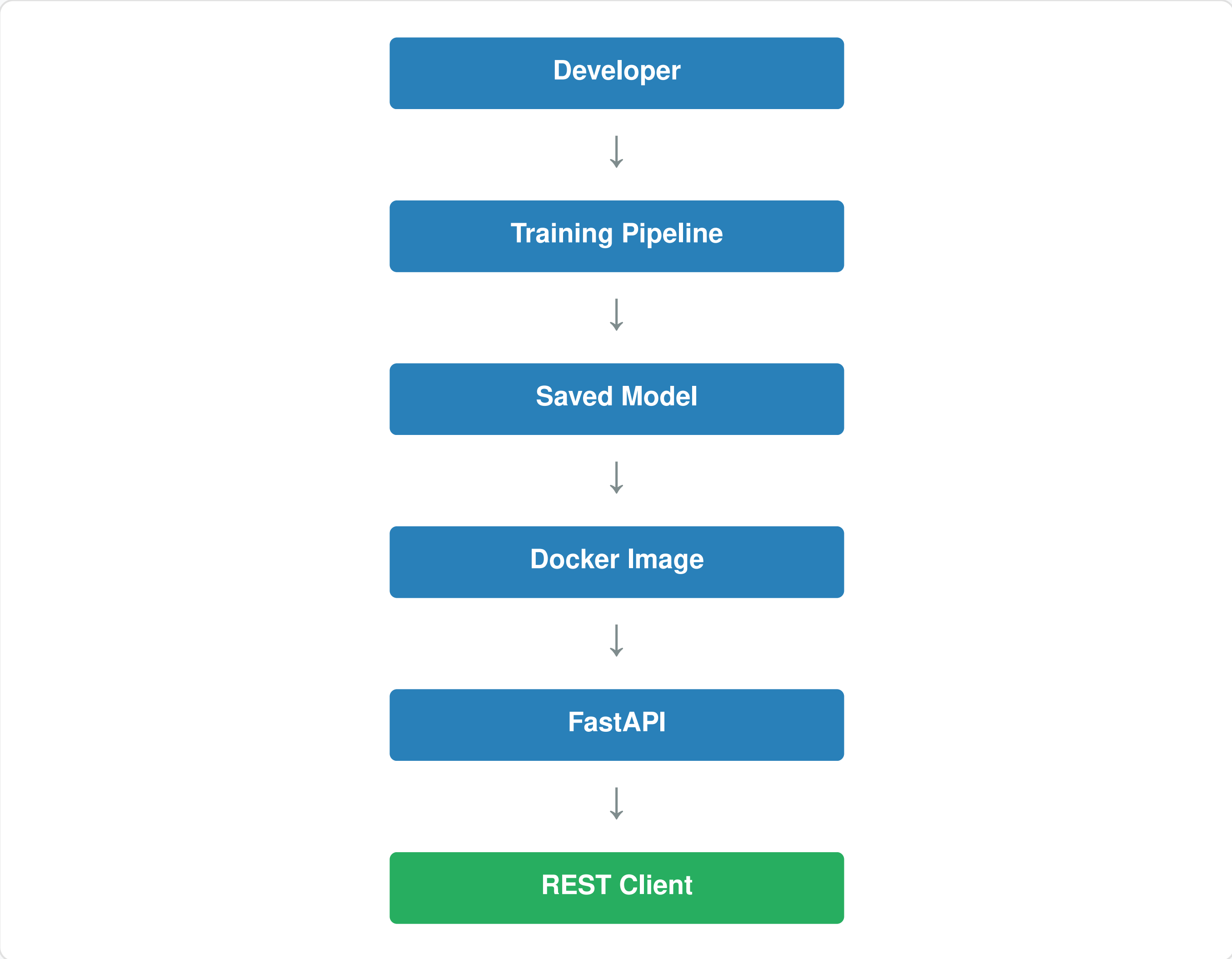
Rank	Model	MAE ↓	RMSE ↓	R^2 ↑	Status
1	**Gradient Boosting**	**1.704**	**1.836**	**0.2664**	**Best**
1	**CatBoost**	**1.704**	**1.836**	**0.2664**	**Best**
2	Lasso Regression	1.708	1.800	0.1340	Excellent
3	Linear Regression	1.823	1.925	0.1820	Baseline
5	Ridge Regression	1.831	1.928	0.1810	Stable

Model Selection: Nineteen regression algorithms were evaluated under an identical validation strategy. Gradient Boosting and CatBoost achieved the lowest MAE (1.704). Gradient Boosting was selected as the default production model due to its strong predictive performance and stable inference behavior.

Furthermore, SHAP-based feature attribution is integrated to improve model interpretability by estimating the contribution of individual features to prediction outputs.

7. Production Deployment & Quality Assurance

Designed to satisfy the functional requirements of the challenge specification, the platform is fully containerized and deployment-ready.



API Performance

Metric	Value
Average Inference Time	~2.36 seconds
Audio Duration Processed	2.84 seconds
Features Extracted	139

Testing Coverage

- Unit Tests
- Feature Extraction Validation
- API Endpoint Testing
- CLI Pipeline Testing
- Model Evaluation Testing
- Configuration Validation
- Automated Regression Tests

Status: 31 / 31 automated tests passing

8. Software Engineering Practices & Reproducibility

Development Workflow

- Git-based version control
- GitHub Actions for automated testing
- Black, Flake8 and Isort for code quality
- Pytest for regression testing
- Docker for reproducible deployment

Reproducibility

The project can be fully reproduced by following these steps:

1. Cloning the repository.
2. Installing dependencies via `requirements.txt`.
3. Running the CLI pipeline.
4. Training the benchmark models.
5. Launching the Dockerized FastAPI service.
6. Running the automated test suite.

Folder Structure

```
src/
 ├── data/
 ├── features/
 ├── models/
 ├── evaluation/
 ├── inference/
 ├── pipeline/
 ├── api/
tests/
configs/
docs/
artifacts/
```

9. Limitations & Future Work

Current Limitations

- Evaluated on the BioDCASE challenge dataset only.
- Inference is optimized strictly for short audio clips.
- Streaming inference is not yet implemented.
- Overall performance remains heavily dependent on audio quality and baseline recording conditions.

Future Work

The final Gradient Boosting model achieved the best validation performance among the evaluated models. Future iterations of this platform will focus on scaling and latency optimization:

- **Transformer-based Audio Models:** Exploring self-supervised representation learning directly on the BioDCASE dataset.
- **ONNX Optimization:** Converting CatBoost/Gradient Boosting models to ONNX runtime for sub-second inference speeds.
- **Real-Time Streaming:** Adapting the FastAPI architecture to accept continuous audio streams for live aviary monitoring dashboards.
- **Cloud Deployment:** Migrating the Dockerized solution to a managed Kubernetes cluster for horizontal scaling during batch inference jobs.

10. Conclusion

This project demonstrates an end-to-end machine learning system for passive acoustic bird population estimation, combining feature engineering, model benchmarking, reproducible experimentation, automated testing, and REST API deployment. Beyond achieving competitive predictive performance, the project emphasizes software engineering practices including modular architecture, containerized deployment, and reproducible workflows.

REFERENCES

[1] Prior analysis and baseline discussion supplied in the challenge materials.

[2] BioDCASE 2026 Bird Counting Challenge specification.